



Software Design

Progress Report No. 4

Rationale of the System

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December 13, 2025

I. Objectives

In this section, the goals in this laboratory are:

- To describe the project objective and its entire features: To develop a functional online booking system with recorded customer history and a login module for Claire Beauty Lounge's "Web-Based Salon Management and Booking System." We aim to ensure there are distinct User Interfaces (UI), each with specific functions and limitations for users, guests, and staff. Only registered users can book appointments, while guests are limited to viewing services.
- To enumerate the project's rationale and its relevant to your field of discipline: This project applies core Computer Engineering principles, specifically Database Management Systems (DBMS) and User Interface (UI) design, to solve real-world operational problems in small-scale retail environments.

II. Methods

Rationale of the System

Many local microbusinesses, specifically beauty salon lounges, rely on manual pen-and-paper methods for recording transactions and plain email or social media messaging apps to receive bookings from customers. This leads to unrecorded transactions and an overall messy booking schedule.

The proposed "Web-Based Salon Management and Booking System" addresses these issues by automating the booking of appointments and recording transactions for each user who booked. This system is essential for microbusiness owners with a small number of employees, as it reduces human error while providing responsive and spontaneous booking demand from a customer and accurate data for each transaction that has happened.

Method

It is essential that the development team of a problem before the team tries to solve the problem. We list all the project deliverables, that is, the user or customer expects to see during project development. Among the deliverables may be:

- Documents
- Demonstration of subsystem

The customer may wish to see the entire procedure in designing the system or the overall output of what the development team can make. All of these are essential parts of the software or system development process.

III. Results

The initial phase of development focused on the database architecture. The figure below visualizes the procedures done regarding data relationships.

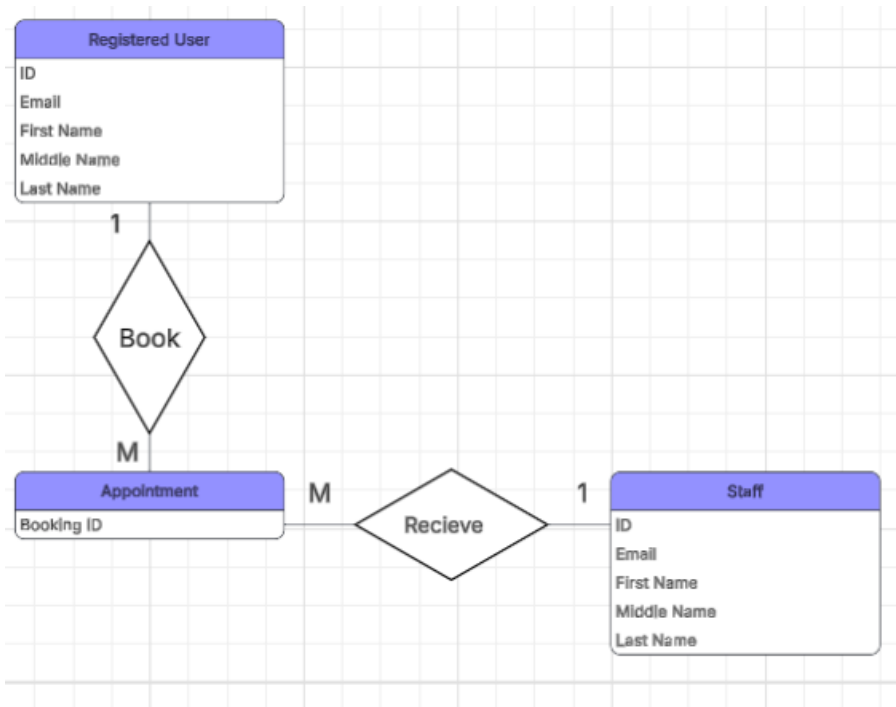


Figure 1 Screenshot of ERD table

As seen in figure 1, 3 entities are made to ensure that a smooth appointment is made. 1:M A registered user can book many appointments and 1:M a staff member can receive many appointments.

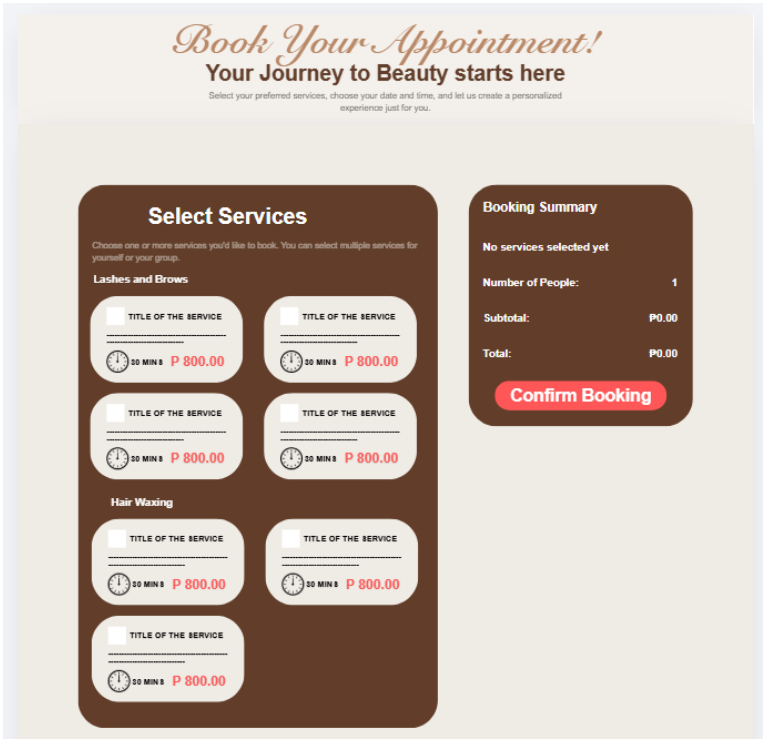


Figure 2 Screenshot of UI made in Canva

2 Your Information

Full Name * Phone Number *

Email Address *

Number of people * Appointment date *

Preferred time *

Special requests *

Figure 3 Screenshot of UI made in Canva

As shown in figure 2 and 3, the team has designed the user interface to be simple for microbusiness employees; the “Confirm Booking” button is prominent.

IV. Conclusion

In this laboratory period, the team successfully conceptualized the architectural design for the "Claire Beauty Lounge Web-Based Salon Management and Booking System." We established the core data relationships necessary for handling appointments through the Entity Relationship Diagram (ERD). Furthermore, the User Interface (UI) was prototyped using Canva to visualize the distinct interaction flows for guests, registered users, and staff. The team concludes that the current design provides a solid foundation to proceed to the coding and implementation phase.

References

- [1] Co Arthur O.. “University of Caloocan City Computer Engineering Department Honor Code,” UCC-CpE Departmental Policies, 2020.